



Lecture 27: Final project presentations

PSYC 51.17: Models of language and communication

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Winter 2026

Today's plan

Final Project Due: March 9, 11:59 PM EST

All final project materials must be submitted **before** today's presentations. Please ensure your notebook, video, and any supplementary materials are uploaded to your GitHub Classroom repository.

Schedule for today

1. **Project presentations** (~5 minutes each + Q&A)
2. **Course wrap-up** and reflections
3. **What comes next** in language and AI

Presentation format

What to cover in your presentation

Section	Time	Content
Problem statement	~1 min	What question did you investigate?
Approach	~1.5 min	What methods, models, or tools did you use?
Results	~1.5 min	What did you find? Show key figures or outputs.
Reflection	~1 min	What worked? What surprised you? What would you do differently?

Tips for a great presentation

- **Show, don't tell:** Demos and visualizations are more compelling than bullet points
- **Be honest:** Negative results and failed approaches are valuable and interesting
- **Connect to course themes:** How does your project relate to concepts we covered?
- **Keep it concise:** 5 minutes goes fast -- practice timing beforehand

Grading criteria

How projects are evaluated

Criterion	Weight	What we are looking for
Technical depth	30%	Meaningful engagement with LLM concepts from the course
Implementation	25%	Working code, clear methodology, reproducibility
Creativity	20%	Original approach, interesting question, novel application
Presentation	15%	Clear communication, good visuals, within time limit
Reflection	10%	Thoughtful analysis of results, limitations, and lessons learned

Reminder

The rubric rewards **depth over breadth**. A focused project with thorough analysis scores higher than a broad project with surface-level treatment.

Presentations begin

While watching presentations

As each team presents, consider:

- What **course concepts** does this project apply?
- What is the most **interesting or surprising** finding?
- What **questions** would you ask the presenters?
- How might you **extend** this work?

Please be a supportive and engaged audience -- ask questions, offer feedback, and celebrate your classmates' work.

Course reflection: our journey

From ELIZA to autonomous agents in 10 weeks

Week	Topic	Key idea
1	Pattern matching (ELIZA, PARRY)	Rules can create the *illusion* of understanding
2	Preprocessing and tokenization	How we represent text as numbers
3--4	Embeddings (LSA → Word2Vec → ELMo)	Meaning lives in vector space
5	Transformer architecture	Attention is all you need
6	BERT and encoder models	Bidirectional understanding
7	GPT and decoder models	Autoregressive generation at scale
8	*(Break)*	
9	Agents, MoE, ethics	Action, efficiency, responsibility

continued...

Course reflection: our journey

From ELIZA to autonomous agents in 10 weeks

Week	Topic	Key idea
10	**Your projects**	Putting it all together

...continued

What we have learned

The big ideas

1. **Language models are compression:** They compress the statistical structure of language into learnable parameters
2. **Scale unlocks capabilities:** Emergent abilities appear at larger model sizes (few-shot learning, reasoning, code generation)
3. **Architecture matters, but data matters more:** The transformer is powerful, but training data quality and quantity drive performance
4. **Alignment is unsolved:** Making models helpful, honest, and harmless remains an open research problem
5. **The field moves fast:** Many concepts we covered were published in the last 2--5 years

What comes next

Frontiers in language modeling (2025 and beyond)

Reasoning models: Systems like o1 and o3 that "think" step-by-step before answering, producing stronger results on math, science, and logic problems.

Multimodal integration: Models that natively process text, images, audio, and video together -- understanding the world through multiple senses simultaneously.

Longer contexts and better memory: Context windows growing to millions of tokens, plus persistent memory across conversations.

Agentic AI: Models that can plan, use tools, write code, and take real-world actions autonomously -- the focus of Lecture 24.

Efficient architectures: State space models, mixture of experts, and new attention variants that reduce compute requirements while maintaining quality.

Open questions for you to carry forward

Questions to consider

- Can language models truly **understand** language, or are they sophisticated pattern matchers? (Recall Lecture 1: Is ChatGPT conscious?)
- How should we handle the **economic disruption** from AI automation?
- What does **responsible development** look like when the technology moves faster than regulation?
- How can we ensure AI benefits are **distributed equitably** across communities and nations?
- What role should **open-source** play in AI development?

These are not just academic questions -- they are decisions you will face in your careers. The perspectives you have developed in this course will help you navigate them thoughtfully.

Thank you

It has been a pleasure

Teaching this course has been a highlight of my year. Your curiosity, engagement, and creative projects have made this class exceptional. I hope the concepts and skills you have developed here serve you well, whether you pursue AI research, build applications, or simply think more critically about the technology shaping our world.

Stay in touch

- Continue discussions on **Discord** -- the server will stay active
- **Office hours** remain open through finals period
- I am always happy to discuss research, graduate school, or career questions

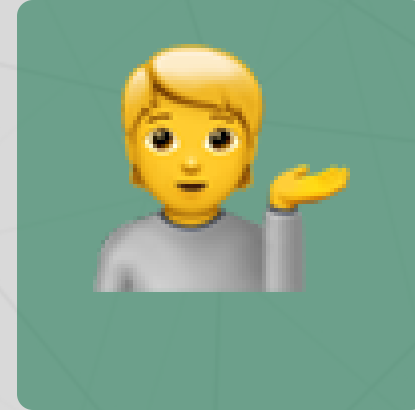
Questions?



Email



Discord



Office hours

That's a wrap!

From ELIZA to autonomous agents -- what a journey. Good luck on your final projects!